



DECISION TABLE

✓ What is a Decision Table?

A **Decision Table** is a **tool** used to model **complicated decision logic** in a simple table format.

Think of it like a **rule-based chart** that tells you:

“If these conditions are met → then do this action.”

It's commonly used in **software engineering**, **business rules**, and **testing**.

★ Basic Parts of a Decision Table

A decision table has 4 parts:

Conditions	Rule 1	Rule 2	...
Condition 1	Y/N	Y/N	...
Condition 2	Y/N	Y/N	...
Actions			
Action 1	X		...
Action 2		X	...

✓ Example: Simple Login System

Let's say we have a login system with these rules:

- Condition 1: Username is valid
- Condition 2: Password is valid

- Action 1: Allow login
- Action 2: Show error

Let's build the table:

Conditions	Rule 1	Rule 2	Rule 3	Rule 4
Username is valid	Y	Y	N	N
Password is valid	Y	N	Y	N
Actions				
Allow login	X			
Show error		X	X	X

💡 Explain:

- Only Rule 1 allows login.
- Every other rule shows an error.

✓ Why Use Decision Tables?

- ✓ Easy to **understand** and explain logic.
- ✓ Helps avoid **missing cases**.
- ✓ Good for **testing** and **automating decisions**.
- ✓ Makes logic **visible** to both tech and non-tech people.

💡 Class Tip: Teach It Like a Story

Say:

Imagine you're a bouncer at a club.

You check two things:

- ID is real (condition 1)

- Person is on the guest list (condition 2)

If both are true → Let them in (Action 1)

Else → Send them away (Action 2)

Make a table with the 4 possibilities and actions, and boom — they get it.

Student Exercise

Have students make a decision table for:

ATM Withdrawal

- Conditions:
 - i. Card is valid
 - ii. PIN is correct
 - iii. Sufficient balance
 - Actions:
 - Allow withdrawal
 - Deny transaction
 - Show insufficient balance
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Summary Cheatsheet:

Section	Purpose
Conditions	Input facts or checks
Rules	All combinations of condition values
Actions	What to do if the rule is true
